Week 2 — DINO from First Principles

Resources

- **DINO Paper:** https://arxiv.org/pdf/2104.14294
- **Additional Explanation:** https://medium.com/@anuj.dutt9/emerging-properties-in-self-supervised-vision-transformers-dino-paper-summary-4c7a6ed68161

Study

Understand the core concepts of DINO:

- Student–Teacher architecture
- EMA/Momentum Teacher
- Multi-crop augmentation
- Teacher centering and sharpening
- DINO loss
- Self-supervised learning without labels
- Emergence of semantic information in ViT attention

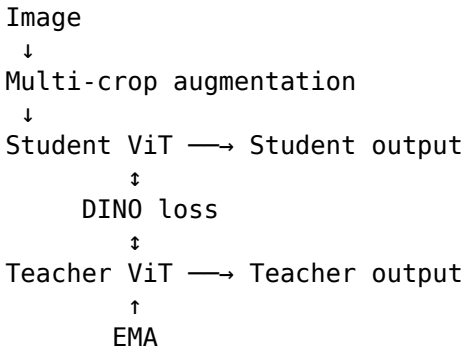
Implementation Task — Mini-DINO

Since the original DINO training setup is computationally expensive, implement a **small-scale DINO version** that can run on free Colab/Kaggle resources.

Use:

- **Dataset:** CIFAR-10, initially with 2,000–5,000 images
- **Model:** Small/Tiny ViT
- **Training:** Short training run (~10–20 epochs)
- **Hardware:** Free Colab/Kaggle GPU

Implement the complete pipeline:



Experiments

- Verify student and teacher outputs during training.
- Verify that the teacher is updated using EMA.
- Plot training loss.

- Extract learned embeddings and visualize them using PCA/t-SNE.
- Visualize ViT attention maps.
- Perform one small ablation, such as **with vs without EMA** or **different crop configurations**.

The goal is **not to reproduce the original DINO training scale**, but to understand and demonstrate the core mechanism through an experiment that can actually run on free GPU resources.

Deliverables

1. **Mini-DINO Colab Notebook**

- Dataset preparation
- Multi-crop augmentation
- Student/Teacher ViT
- DINO loss
- EMA update
- Training
- Embedding and attention visualization

2. **Short Week 2 Report**

- DINO architecture and intuition
- Student–Teacher mechanism
- EMA, multi-crop, centering and sharpening
- Experimental observations
- Ablation results

3. Upload the notebook, report, and results to GitHub under:

```
Week-2/
├── DINO_Mini_Implementation.ipynb
├── Week2_Report.pdf
└── results/
```

Main objective: Implement a small but complete DINO pipeline and understand how self-distillation can learn useful visual representations without labeled data.